Fiscal Year:	FY 2012	**Task Last Updated:**	FY 01/05/2012
PI Name:	Blakely, Eleanor A Ph.D.		
Project Title:	Harderian Gland Tumorigenesis: Low-Dose-, Low-Dose-Rate-, and LET-Response		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiation NSCOR/Virtual NSCOR NNJ10ZSA002N
Start Date:	12/20/2010	**End Date:**	12/30/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Gap change to delete Cancer 05, per PI (Ed., 12/17/12)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chang, Polly (SRI International)		
Grant/Contract No.:	NNJ11HA94I		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Increased cancer risk remains a primary concern for travel into deep space and may preclude manned missions to Mars due to large uncertainties that currently exist in estimating cancer risk from the spectrum of radiations found in space with available human epidemiological radiation-induced cancer data. Existing data on human risk of cancer from exposures to X- and gamma-rays must be scaled to the many types and fluences of radiations found in space using radiation quality factors and dose-rate modification factors, and assuming linearity of response since the shapes of the dose-responses at low doses below 100 mSv are unknown. The goal of this proposal is to reduce uncertainties in the RBE/LET relationship for particle radiation carcinogenesis. Our approach is to use the existing data on Harderian gland tumorigenesis as reference, and obtain Harderian gland tumor data for heavy ions beams not previously reported to fill the gaps in the LET range from below 25 keV/micrometer and up to approximately 193 keV/micrometer to improve our		

	understanding of the dose-response curve at low doses and to test for deviations from linearity. The hypothesis to be tested is that filling in these data gaps will delineate more comprehensively the space radiation risk of space radiation induced tumorigenesis, in particular to provide data that will allow us to test for a role of targeted versus non-targeted effects in cancer risk. We aim to investigate the effects of a broader range of radiation qualities, doses and dose rates not previously reported on the development of Harderian gland and also lung tumors in ~100-120 day old female CB6F1 mice 16 months after exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The successful completion of the proposed work will fill a significant gap in Harderian tumor response data at low fluences of Silicon or Titanium particle beams to fill in the gaps in the LET range from below 25 keV/micrometer and up to approximately 193 keV/micrometer to improve our understanding of the dose-response curve at low doses relevant to space travel and to test for deviations from linearity. The data should contribute to decreasing the current uncertainty in estimating radiation risk of cancer due to exposures in space travel.
Task Progress:	We participated in the NSRL Spring run in 2011. Ms. Lori Chappell, our collaborator at NASA, has performed power calculations to determine the number of animals needed for each experiment in order to obtain results that are statistically meaningful at the relevant low particle radiation doses specified. CB6F1 hybrid mice were purpose-bred by a certified vendor and shipped to BLAF (Brookhaven Lab Animal Facility) for quarantine and acclimation prior to scheduled irradiation exposure time. Animals were multiply-housed and randomized by weight and birthdates into dose groups. To be consistent with the previous studies, animals were between the ages of 100 – 120 days old at the time of exposure. Animals were briefly restrained for less than 10 min in approved aerated plastic holders and whole-body irradiated. During irradiation, animals on the beamline were remotely monitored outside the cave using a video camera system that is mounted in the irradiation cave. All animals appeared to be resting quietly during the short irradiation period. Unirradiated controls were also loaded into the cubic holders but not irradiated. Since the radiation exposure doses are small, the exposure times were less than 1 min each. After exposure, animals were unloaded to their housing cage with their cage mates and transported back to BLAF for temporary housing. Animals were shipped to the LBNL Animal Colony Facility (ACF) by an authorized animal courier prequalified by both BNL and LBNL. Animals will be maintained at LBNL for 16 months. During this period, animals will be monitored on a daily basis to assess any physical or behavioral changes that might indicate that an animal is experiencing distress, discomfort, pain or injury per institutional animal care guidelines. Mice will be weighed once a month to help monitor health. If a downward trend in the weight of the mice is noted, they will be weighed more frequently, twice a month or even once a week. Any abnormal observations will be reported to the animal core facility manager for evaluation, with potential follow-up by the attending veterinarian. At the end of 16 months, Harderian glands, the spleen, portions of two mammary glands and any tumors from each animal will be collected and processed for histopathology analysis. In some cases, only a portion of the gland will be fixed, while the other portion will be flash frozen for molecular profiling. As a consortium member of the LUNG NSCOR/VIRTUAL NSCOR, we have actively participated in team discussions and are developing collaborations for tissue sharing, access to clinical expertise, and developing experimental plans to advance our understanding of molecular mechanisms specifically underlying particle-induced carcinogenesis. Several talks and posters of our experimental design and progress to date have been presented at meetings. Manuscripts reporting our results will be written following our completion of the experiments in progress.
Bibliography Type:	Description: (Last Updated: 02/12/2026)
Abstracts for Journals and Proceedings	Blakely EA. "What makes particle radiation so effective?" 35th Lauriston S. Taylor Lecture Award. 2011 NCRP Annual Meeting, Bethesda, MD, March 7-8, 2011. Program Presentations No. 47 - Scientific and Policy Challenges of Particle Radiations in Medical Therapy and Space Missions. Bethesda, Md. : National Council on Radiation Protection and Measurements, 2011. , Mar-2011
Abstracts for Journals and Proceedings	Blakely EA. "Cataract and Cancer: Late Effects of Particle Radiation." Presented at the International Symposium for Radiation Research and Medical Physics, Shanghai, China, May 30-June 2, 2011. International Symposium for Radiation Research and Medical Physics, Shanghai, China, May 30-June 2, 2011. , Jun-2011
Abstracts for Journals and Proceedings	Chang PY, Blakely EA. "Harderian Gland Tumorigenesis: Low-Dose-, Low-Dose-Rate- and LET-Response." Presented at 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. p. 241. , Sep-2011
Abstracts for Journals and Proceedings	Blakely EA, Chang PY. "Harderian Gland Tumorigenesis: Low-Dose-, Low-Dose-Rate- and LET-Response." Presented at the 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7012.pdf , Sep-2011
Awards	Blakely EA. "35th Lauriston S. Taylor Lecture Award for 'What makes particle radiation so effective?'. 2011 NCRP Annual Meeting, Bethesda, MD, March 7-8, 2011." Mar-2011